

Tutorial 3

Introduction to Linux and bash commands

Overview

Linux

- Introduction
- WSL
- VirtualBox
- MinGW

Bash commands

- Basic Commands
- Case Study

Introduction To Linux



Linux

- Linux is a free, open-source operating system (OS) first released by Linus Torvalds in 1991.
- An operating system (OS) is software that manages a computer's hardware and software resources, providing a platform for applications to run.
- It's based on the Unix operating system and is known for its robustness, security, and flexibility.
- Since Linux is open-source, its source code can be modified and distributed by anyone. There are various different distributions of Linux.
- Some popular distributions include: Mint and Ubuntu
- You can interact with Linux through the shell (terminal) or through the GUI (graphical user interface)

WSL (Windows Subsystem For Linux)

How to install it:

- Search for the app “**Windows Powershell**”
- Right click and press “**Run as administrator**”
- Write the command: `Enable-WindowsOptionalFeature -Online -FeatureName Microsoft-Windows-Subsystem-Linux`
- Write the command: `dism.exe /online /enable-feature /featurename:VirtualMachinePlatform /all /norestart`
- Write the command: `dism.exe /online /enable-feature /featurename:Microsoft-Windows-Subsystem-Linux /all /norestart`
- A terminal window will open up, write in the command: `wsl --install`
- Your computer will ask to restart, restart your computer
- Search for the app “wsl”
- Go to Microsoft Store and search wsl. Download the newest Ubuntu version.
- It will ask you to enter a username and password for your wsl machine

Link for reference: [Installing WSL](#)

Alternatives: VirtualBox

VirtualBox is a free, open-source software created by Oracle that allows users to run multiple operating systems simultaneously on a single physical machine, using a concept known as virtualization.

Cross-Platform: Works on Windows, macOS, Linux, and Solaris hosts.

Step 1: Install VirtualBox

- Download VirtualBox: Go to the VirtualBox website and download the version compatible with your host operating system (Windows or macOS).
- Install VirtualBox: Run the installer and follow the prompts to install VirtualBox on your system.

Step 2: Download a Linux Distribution

- Choose a Linux distribution (distro) such as Ubuntu, Fedora, or Debian. For beginners, Ubuntu is highly recommended due to its user-friendly interface and extensive documentation.
- Download the ISO file for your chosen distro. For Ubuntu, you can download it from Ubuntu's official website.

Step 3: Create a New Virtual Machine (VM)

- Open VirtualBox and click on the "New" button to create a new VM.
- Name your VM and choose "Linux" as the type and the version that matches your downloaded distro (e.g., Ubuntu (64-bit)).
- Allocate RAM: Assign a portion of your system's memory to the VM. For general use, 2048 MB (2 GB) is a good starting point.
- Create a Virtual Hard Disk: Follow the prompts to create a virtual hard drive for your Linux installation. A dynamically allocated file is usually fine.

Step 4: Install Linux on the VM

- Mount the Linux ISO: With the VM selected, click on "Settings" -> "Storage" -> "Empty" under the Controller: IDE, then click on the disk icon next to "Optical Drive" and select "Choose a disk file." Navigate to and select the Linux ISO you downloaded.
- Start the VM and Install Linux: Click "Start" to boot your VM. You should see the Linux installation menu. Proceed with the installation process as you would on a physical machine.

Please look for videos to help you if you struggle: [Windows 10](#), [Windows 11](#), [MacOS](#)

MinGW (Minimalist GNU for Windows)

An open-source tool that lets you use GCC (a program to run your C or C++ code) to create Windows apps and run C++ on Windows without needing Microsoft's own proprietary tools.

Note:

- MinGW uses the Windows approach to memory management, which can be different from Unix/Linux systems
- This might mean that you don't pick up errors on your machine that will be picked up by FF
- If you choose to use MinGW be sure to test your code on the lab computers that run Linux to make sure that there is no unexpected behaviour.

Instructions to install: <https://www.youtube.com/watch?v=FEeFG9OR-QU>

Introduction to Bash Commands

Bash is an enhanced version of the initial Linux shell (terminal)

Basic Bash commands allow users to:

- Navigate through a system,
- Effectively manage files,
- Manage directories,
- and Archive/Unzip files

File and Directory Operations

- **ls Command**
 - The ls command lists all files and directories in the current directory
 - Example:
- **cd Command**
 - The cd command changes the current directory to the directory specified as an argument to the command
 - You can use “**cd** . . / ” to move back one directory
 - You can use “**cd** ~ ” to move you to your home directory
 - Example:
- **pwd Command**
 - The pwd command prints the path of the current working directory.
 - Example:

File Manipulation

- **touch** Command
 - The touch command creates a new, empty file or updates the timestamp of an existing file.
 - Example:
- **rm** Command
 - The rm command removes files or directories to free up storage space and organize the file system
 - Example:
- **mkdir** Command
 - The mkdir command creates a new, empty directory.
 - Example:
- **rmdir** Command
 - The rmdir command removes only empty directories via the terminal
 - Example:
- **nano** Command
 - The nano command allows you to edit a file.
- **Others:** cat (concatenate), cp (copy), mv (move)

Archiving and Compression

- **tar** Command
 - The tar command is an archiving tool for creating, extracting, and manipulating archive files called tarballs
- **gzip/gunzip** Command
 - **gzip** command is a file compression utility used to reduce the size of files while preserving the content
 - **gunzip** command is a utility for decompressing files that have been compressed using the **gzip**
- **zip** Command
 - file compression utility that uses a different algorithm to tar and gzip
 - Zip provides capabilities roughly equivalent to the combination of tar and gzip
 - Not pre-installed on linux.
- **Installing Packages in Linux:** eg. zip
 - `sudo apt-get update`
 - `sudo apt-install [package-name]`
 - Eg: `sudo apt-get install zip`

Case Study

You are enrolled in COS 132 Module at the University of Pretoria and are given a practical exam.

The exam requires you to perform the following tasks on a linux run machine using the command line:

- Given the practical 2 compressed folder
- Extract the archive
- Update task1.cpp timestamp, delete any file that's not a .cpp file type, add an empty text file (named readme) to a new folder (named info) in the current directory
- submit a compressed archive containing the remaining files/folders

Demonstration

- Navigate to practical 2: **cd** and **ls**
- Extract the archive: **zip**
- Move into folder and list files: **cd** and **ls**
- Update task1.cpp timestamp: **touch**
- List the files to see the updated timestamp. Use the **-l** flag with **ls** to use long listing
- Delete any file that's not a .cpp file type: **rm**
- Add an empty text file (named readme) using **touch** to a new folder (named info) using **mkdir** in the current directory
- Edit the readme.txt to contain "Hello world" using **nano**
- Submit a **zip** file containing task1.cpp using **zip**. The zip should be named XXXXXXXXXX.zip where XXXXXXXXX is your student number